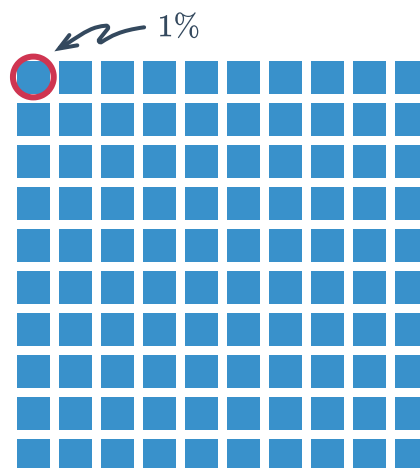


Addition and subtraction in percentages change

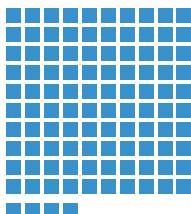


- 1 Here is a grid of 100 blue squares. Each small square represents 1% of the total:

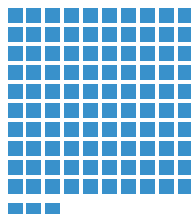


- a Choose the image that represents 3% more than the original grid:

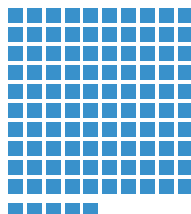
A



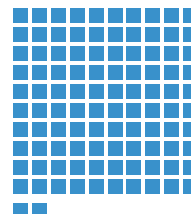
B



C

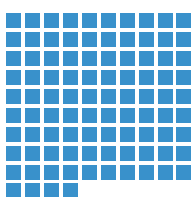


D

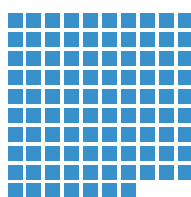


- b Describe what was done to the original grid to increase it by 3%.
- c Write an expression to represent the 3% increase in squares.
- d Choose the image that represents 6% less than the original grid:

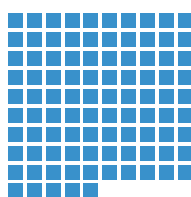
A



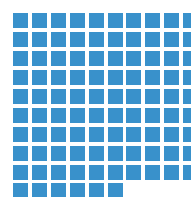
B



C



D



- e Describe what was done to the original grid to decrease it by 6%.
- f Write an expression to represent the 6% decrease in squares.

- 2 Bob wants to decrease 110 by 60%.

- a Calculate 60% of 110.
- b Should Bob add or subtract the amount from part (a) from 110? Explain your answer.



- 3** Lucy wants to increase 70 by 60%.
- a** Calculate 60% of 70.
 - b** Should Lucy add or subtract the amount from part (a) from 70? Explain your answer.
 - c** Hence, increase 70 by 60%.

Percentage increase

4 Increase:

- | | |
|------------------------|---------------------------|
| a 80 by 5%. | b 120 by 10%. |
| c 72 by 40%. | d 320 by 25%. |
| e \$140 by 9%. | f \$30 000 by 20%. |
| g 80 km by 15%. | h 400 mL by 35%. |

5 Abed wants to increase 1500 by 30%.

- a** Calculate 30% of 1500 and add the answer to 1500.
- b** Calculate 130% of 1500.
- c** Are your answers to (a) and (b) the same? Explain why this is.

6 Tracy wants to increase 80 by 30%.

- a** What single percentage should she multiply 80 by?
- b** Explain how you got your percentage in part (a).

7 What single percentage should a quantity be multiplied by in order to increase the quantity by:

- | | | | |
|--------------|---------------|--------------|---------------|
| a 17% | b 5% | c 40% | d 9.4% |
| e 33% | f 127% | g 38% | h 212% |

8 Evaluate the following using a single percentage:

- | | |
|-------------------------------|------------------------------|
| a Increase 80 by 60% | b Increase 400 by 45% |
| c Increase 900 by 8% | d Increase 80 by 40% |
| e Increase 1200 by 45% | f Increase 1200 by 8% |

- 9 A bag of rice weighs 120 kg. The company  makes these bags decides to increase their weight by 45%. Find the new weight of the bag.
-

Percentage decrease

10 Decrease:

- | | |
|------------------|--------------------|
| a 70 by 8%. | b 260 by 10%. |
| c 96 by 60%. | d 250 by 20%. |
| e \$480 by 18%. | f \$55 000 by 45%. |
| g 200 km by 22%. | h 650 mL by 30%. |

11 Derek wants to decrease 120 by 70%.

- a What single percentage should he multiply 120 by?
- b Explain how you got your percentage in part (a).

 12 What single percentage should a quantity be multiplied by in order to decrease the quantity by:

- | | | | |
|-------|-------|-------|-------|
| a 17% | b 5% | c 40% | d 9% |
| e 32% | f 52% | g 38% | h 65% |

13 Evaluate the following using a single percentage:

- | | |
|------------------------|-----------------------|
| a Decrease 80 by 60% | b Decrease 400 by 45% |
| c Decrease 900 by 8% | d Decrease 80 by 40% |
| e Decrease 1200 by 45% | f Decrease 1200 by 8% |

14 A discus thrower has a personal best of 60 m. In their most recent competition they fall short of their personal best by 30%. Find the distance of their throw in this competition.

Percentage change

15 A percentage increase was applied to the number 60 in order to get 108.

- a Find how much was added to 60 to get 108.
- b Express the increase as a percentage of 60.



- 16** A percentage increase was applied to the number 180 in order to get 207.
- Find how much was added to 180 to get 207.
 - Express the increase as a percentage of 180.
- 17** A percentage decrease was applied to the number 90 in order to get 27.
- Find how much was subtracted from 90 to get 27.
 - Express the decrease as a percentage of 90.
- 18** A percentage decrease was applied to the number 180 in order to get 117.
- Find how much was subtracted from 180 to get 117.
 - Express the decrease as a percentage of 180.
- 19** In training for her next marathon, Sally increased her practise route from 7000 m to 7910 m. What percentage has Sally increased the distance of her practice route by?
- 20** Luke's record for juggling a football was 70 keep ups. After buying some new football boots, this increased to 112. Express this increase as a percentage of the original record.
- 21** Sharon's daily pay was \$150. When she began working shorter hours in order to study, her pay decreased to \$90 per day. Express this decrease as a percentage of her original pay.
- 22** Jimmy had a bowling high score of 60 points. After practising for a week, this increased to 93. Express this increase as a percentage of his original high score.
- 23** Sophia is able to write 220 words in a minute. After sustaining a thumb injury, this decreased to 143 words a minute. Express this decrease as a percentage of her original rate.
- 24** During the editing phase when making a movie, the director ordered that the movie be cut down from 200 minutes to 124 minutes. What percentage did the movie's duration decrease by from its original length?